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City of Cadillac, Water Resources
200 North Lake Street
Cadillac, MI 49601

Attn : Ms. Amy Vail

KAR Project No. : 602902

Date Reported : 03/29/16

Date Activated : 03/15/16

Date Due : 03/29/16

Date Validated : 03/29/16

Project

Description : Analysis of one Spring Biosolids sample.

Dear Client,

Your laboratory data is presented to you in this report. Unless otherwise stated under the "Comments" heading, all tests were performed within the maximum allowable holding times, have met or exceeded QC requirements and the result represents the sample as it was received. If a sample was identified as drinking water under the Safe Drinking Water Act, the "Comments" column may also contain federal drinking water information including MCL which is the Maximum Contaminant Level set by USEPA. Values enclosed in brackets ([]) are Secondary MCL's and are non-enforceable guidelines for aesthetic quality.

If you wish to contact us about this work please mention KAR Project No. 602902. To arrange additional sampling or testing please contact our Client Services Department. If you have any questions regarding quality assurance please contact us.

Thank you for the opportunity to serve you. Please do not hesitate to call if we can provide additional assistance.

Respectfully submitted,

A handwritten signature in black ink that reads 'David R. Alkema'.

David R. Alkema
Laboratory Manager

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LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **602902**

Attest: 
David R. Alkema, Lab Manager

Date Reported: **03/29/16**

Project

Description : *Analysis of one Spring Biosolids sample.*

Sample ID : **"Digester #4"**

Sampled By :

Date Received : **03/15/16**

Sample Date : **03/14/16**

Sample Type : **sludge**

Sample Time : **1400**

KAR Sample No. : **602902-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prep. Hg	Completed		EPA 7471A	03/21/16	STC	
Prep. metals	Completed		EPA 3050B	03/16/16	STC	
Aluminum, total	3310	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 36
Arsenic, total	26.1	mg/kg total solids	EPA 200.8	03/23/16	NHM	Detection limit = 14
Cadmium, total	<7.1	mg/kg total solids	EPA 200.8	03/23/16	NHM	
Calcium, total	27,700	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 36
Chromium, total	85.7	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 3.6
Copper, total	843	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 7.2
Lead, total, by ICP	55.0	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 17.8
Magnesium, total	4600	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 36
Mercury, total	1.76	mg/kg total solids	EPA 7471A	03/21/16	STC	Detection limit = 0.15
Molybdenum, total	23.1	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 7.2
Nickel, total	28.5	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 7.2
Potassium, total	4100	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 36
Selenium, total	<7.1	mg/kg total solids	EPA 200.8	03/23/16	NHM	
Silver, total	<1.8	mg/kg total solids	EPA 6010B	03/22/16	JHB	
Sodium, total	5960	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 360
Zinc, total	1330	mg/kg total solids	EPA 6010B	03/18/16	JHB	Detection limit = 7.2
TCLP extraction	Completed		EPA 1311	03/17/16	JWW	
TCLP report	See comment			03/29/16	CLK	This material does not exhibit the Toxicity Characteristic with respect to the requested TC parameters.
ZHE extraction	Completed		EPA 1311	03/17/16	JWW	
Chloride	7890	mg/kg total solids	EPA 300.0A	03/23/16	ALK	Detection limit = 290
Nitrogen, ammonia	53,700	mg/kg total solids	EPA 350.1	03/16/16	JWW	Detection limit = 1200
Nitrogen, nitrate	<29	mg/kg total solids	EPA 300.0A	03/23/16	ALK	
Nitrogen, total kjeldahl	90,800	mg/kg total solids	EPA 351.2	03/22/16	ALK	Detection limit = 2500
Phosphorus, total (as P)	41,400	mg/kg total solids	SM 4500-P B5,E	03/21/16	EIF	Detection limit = 12,000
Sulfate	3840	mg/kg total solids	EPA 300.0A	03/23/16	ALK	Detection limit = 580
Waste pH	7.8	S.U.	EPA 9040C	03/29/16	AJK	
Solids, total	1.75	% by weight	EPA 160.3	03/17/16	EIF	Detection limit = 0.01
Solids, volatile	59.00	% of total solids	EPA 160.4	03/17/16	EIF	Detection limit = 0.01
Prior. Poll. volatiles	See below		EPA 8260	03/17/16	JAR	
Prep. VOA	Completed		EPA 5030	03/17/16	JAR	
1,1,1-Trichloroethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,1,2,2-Tetrachloroethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,1,2-Trichloroethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,1-Dichloroethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,1-Dichloroethene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,2-Dichlorobenzene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,2-Dichloroethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,2-Dichloropropane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
1,3-Dichlorobenzene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **602902**

Attest: 
David R. Alkema, Lab Manager

Date Reported: **03/29/16**

Project

Description : **Analysis of one Spring Biosolids sample.**

Sample ID : **"Digester #4"**

Sampled By :

Date Received : **03/15/16**

Sample Date : **03/14/16**

Sample Type : **sludge**

Sample Time : **1400**

KAR Sample No. : **602902-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
1,4-Dichlorobenzene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
2-Chloroethylvinyl ether	<30,000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Acrolein	<60,000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Acrylonitrile	<6000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Bromodichloromethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Bromoform	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Bromomethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Carbon tetrachloride	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Chlorobenzene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Chloroethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Chloroform	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Chloromethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Cis-1,3-Dichloropropene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Dibromochloromethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Ethylbenzene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Methylene chloride	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Tetrachloroethene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Toluene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Trans-1,2-Dichloroethene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Trans-1,3-Dichloropropene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Trichloroethene	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Trichlorofluoromethane	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Vinyl chloride	<3000	ug/kg total solids	EPA 8260	03/17/16	JAR	
Prior. Poll. acids	See below		EPA 8270	03/16/16	GMB	
Prior. Poll. base-neutrals	See below		EPA 8270	03/16/16	GMB	
Prep. SV Acid/BN	Completed		EPA 3545	03/16/16	GMB	
1,2,4-Trichlorobenzene 8270	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
1,2-Dichlorobenzene by 8270	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
1,2-Diphenylhydrazine	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
1,3-Dichlorobenzene by 8270	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
1,4-Dichlorobenzene by 8270	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,3,7,8-TCDD by 8270	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,4,6-Trichlorophenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,4-Dichlorophenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,4-Dimethylphenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,4-Dinitrophenol	<38,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,4-Dinitrotoluene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2,6-Dinitrotoluene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2-Chloronaphthalene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2-Chlorophenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2-Methyl-4,6-dinitrophenol	<38,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
2-Nitrophenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	

KAR Laboratories, Inc.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **602902**

Attest: 
David R. Alkema, Lab Manager

Date Reported: **03/29/16**

Project

Description : *Analysis of one Spring Biosolids sample.*

Sample ID : **"Digester #4"**

Sampled By :

Date Received : **03/15/16**

Sample Date : **03/14/16**

Sample Type : **sludge**

Sample Time : **1400**

KAR Sample No. : **602902-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
3,3'-Dichlorobenzidine	<38,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
4-Bromophenyl phenyl ether	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
4-Chloro-3-methylphenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
4-Chlorophenyl phenyl ether	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
4-Nitrophenol	<38,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Acenaphthene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Acenaphthylene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Anthracene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Benzidine	<94,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Benzo(a)anthracene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Benzo(a)pyrene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Benzo(b)fluoranthene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Benzo(ghi)perylene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Benzo(k)fluoranthene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Bis(2-chloroethoxy)methane	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Bis(2-chloroethyl)ether	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Bis(2-chloroisopropyl)ether	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Bis(2-ethylhexyl)phthalate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Butylbenzyl phthalate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Chrysene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Di-n-butyl phthalate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Di-n-octyl phthalate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Dibenzo(ah)anthracene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Diethyl phthalate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Dimethyl phthalate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Fluoranthene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Fluorene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Hexachlorobenzene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Hexachlorobutadiene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Hexachlorocyclopentadiene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Hexachloroethane	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Indeno(123cd)pyrene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Isophorone	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
N-Nitrosodi-n-propylamine	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
N-Nitrosodimethylamine	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
N-Nitrosodiphenylamine	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Naphthalene by Method 8270	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Nitrobenzene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Pentachlorophenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Phenanthrene	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Phenol	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Pvrene	<19,000	ua/ka total solids	EPA 8270	03/16/16	GMB	

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **602902**

Attest: 
David R. Alkema, Lab Manager

Date Reported: **03/29/16**

Project

Description : *Analysis of one Spring Biosolids sample.*

Sample ID : **"Digester #4"**

Sampled By :

Date Received : **03/15/16**

Sample Date : **03/14/16**

Sample Type : **sludge**

Sample Time : **1400**

KAR Sample No. : **602902-01**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
Prior. Poll. pest./PCB's	See below		EPA 8270	03/16/16	GMB	
4,4'-DDD	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
4,4'-DDE	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
4,4'-DDT	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Aldrin	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Alpha-BHC	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Beta-BHC	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Chlordane	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Delta-BHC	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Dieldrin	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Endosulfan I	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Endosulfan II	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Endosulfan sulfate	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Endrin	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Endrin aldehyde	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Gamma-BHC	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Heptachlor	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Heptachlor epoxide	<19,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1016 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1221 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1232 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1242 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1248 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1254 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
PCB Aroclor 1260 by 8270	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	
Toxaphene	<190,000	ug/kg total solids	EPA 8270	03/16/16	GMB	

LABORATORY DETAIL REPORT

Client: **City of Cadillac, Water Resources**

KAR Project No. : **602902**

Date Reported: **03/29/16**

Attest: 
David R. Alkema, Lab Manager

Project

Description : **Analysis of one Spring Biosolids sample.**

Sample ID : **TCLP Leachate of Digester #4**

Sampled By :

Date Received : **03/15/16**

Sample Date :

Sample Type : **TCLP**

Sample Time :

KAR Sample No. : **602902-01T**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Metals	Completed		EPA 200.8	03/21/16	NHM	
Prep. Hg	Completed		EPA 7470A	03/28/16	STC	
Prep. metals	Completed		EPA 200.7/8	03/18/16	STC	
Arsenic, total, by ICP	<0.5	mg/L	EPA 200.8	03/21/16	NHM	TC regulatory limit is 5.0 mg/L.
Barium, total	1.23	mg/L	EPA 200.8	03/21/16	NHM	TC regulatory limit is 100 mg/L.
Cadmium, total	<0.025	mg/L	EPA 200.8	03/21/16	NHM	TC regulatory limit is 1.0 mg/L.
Chromium, total	<0.05	mg/L	EPA 200.8	03/21/16	NHM	TC regulatory limit is 5.0 mg/L.
Copper, total	<0.1	mg/L	EPA 200.8	03/21/16	NHM	
Lead, total, by ICP	<0.25	mg/L	EPA 200.8	03/21/16	NHM	TC regulatory limit is 5.0 mg/L.
Mercury, total	<0.05	mg/L	EPA 7470A	03/28/16	STC	TC regulatory limit is 0.2 mg/L.
Selenium, total, by ICP	<0.5	mg/L	EPA 200.8	03/21/16	NHM	TC regulatory limit is 1.0 mg/L.
Silver, total	<0.025	mg/L	EPA 200.7	03/22/16	JHB	TC regulatory limit is 5.0 mg/L.
Zinc, total	0.97	mg/L	EPA 200.8	03/21/16	NHM	
TC Semi-Volatiles	See below		EPA 8270	03/21/16	GMB	
Prep. SV Acid/BN	Completed		EPA 3510	03/18/16	GMB	
Prep. herbicides	Completed		EPA 8151	03/18/16	GMB	
2,4,5-Trichlorophenol	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 400 mg/L.
2,4,6-Trichlorophenol	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 2.0 mg/L.
2,4-Dinitrotoluene	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 0.13 mg/L.
Cresols	<0.2	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 200 mg/L.
Hexachlorobenzene	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 0.13 mg/L.
Hexachlorobutadiene	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 0.5 mg/L.
Hexachloroethane	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 3.0 mg/L.
Nitrobenzene	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 2.0 mg/L.
Pentachlorophenol	<0.1	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 100 mg/L.
Pyridine	<0.2	mg/L	EPA 8270	03/21/16	GMB	TC regulatory limit is 5.0 mg/L.
TC Pesticides by ECD	See below		EPA 8081	03/21/16	GMB	
Prep. ECD	Completed		EPA 3510	03/18/16	GMB	
Chlordane	<0.02	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 0.03 mg/L.
Endrin	<0.004	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 0.02 mg/L.
Heptachlor	<0.004	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 0.008 mg/L.
Heptachlor epoxide	<0.004	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 0.008 mg/L.
Lindane by ECD	<0.004	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 0.4 mg/L.
Methoxychlor	<0.02	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 10.0 mg/L.
Toxaphene	<0.04	mg/L	EPA 8081	03/21/16	GMB	TC regulatory limit is 0.5 mg/L.
TC Herbicides	See below		EPA 8151	03/22/16	GMB	
2,4,5-TP (Silvex)	<0.2	mg/L	EPA 8151	03/22/16	GMB	TC regulatory limit is 1.0 mg/L.
2,4-D	<0.2	mg/L	EPA 8151	03/22/16	GMB	TC regulatory limit is 10.0 mg/L.

LABORATORY DETAIL REPORT

Client: *City of Cadillac, Water Resources*

KAR Project No. : **602902**

Date Reported: **03/29/16**

Attest: 
David R. Alkema, Lab Manager

Project

Description : *Analysis of one Spring Biosolids sample.*

Sample ID : **ZHE Leachate of Digester #4**

Sampled By :

Date Received : **03/15/16**

Sample Date :

Sample Type : **ZHE**

Sample Time :

KAR Sample No. : **602902-01Z**

Test	Result	Units of Measure	Method	Analyzed	Analyst	Comments
TC Volatiles	See below		EPA 8260	03/22/16	JAR	
Prep, VOA	Completed		EPA 5030	03/22/16	JAR	
1,1-Dichloroethene	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.7 mg/L.
1,2-Dichloroethane	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.5 mg/L.
1,4-Dichlorobenzene	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 7.5 mg/L.
Benzene	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.5 mg/L.
Carbon tetrachloride	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.5 mg/L.
Chlorobenzene	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 100 mg/L.
Chloroform	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 6.0 mg/L.
Methyl ethyl ketone	<2	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 200 mg/L.
Tetrachloroethene	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.7 mg/L.
Trichloroethene	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.5 mg/L.
Vinyl chloride	<0.1	mg/L	EPA 8260	03/22/16	JAR	TC regulatory limit is 0.2 mg/L.

[illegible]